RISHI THAKUR

DATA ANALYST

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PROFESSIONAL SUMMARY

Data Analyst with a quantitative foundation (M.Tech CSE, GATE CS qualified) and expertise in SQL data warehousing, dimensional modeling, and business intelligence. Experienced architecting Kimball star schemas, authoring dbt pipelines with 213 automated tests, and building interactive Power BI and Metabase dashboards. Proficient in Python (Pandas, Scikit-learn) for exploratory data analysis, statistical modeling, and translating operational data into executive insights.

SKILLS

- **Core Analytics & Programming:** Python (Pandas, NumPy, Scikit-learn), Exploratory Data Analysis (EDA), Statistical Modeling, Hypothesis Testing, Outlier Detection, Data Cleansing
- **Databases & Advanced SQL:** Advanced SQL (MySQL, PostgreSQL, MSSQL, T-SQL) — Joins, CTEs, Window Functions, Indexing, Query Optimization, View Creation
- **BI, Data Warehousing & Modeling:** Microsoft Power BI (DAX, Power Query), Metabase, Kimball Star Schema, Dimensional Modeling, dbt Core, Apache Airflow, Docker, Advanced Excel
- **Communication & Spoken Skills:** Executive KPI Presentations, Cross-Functional Communication, Visual Data Storytelling, Requirements Elicitation, Technical Documentation

KEY PROJECTS

Production E-Commerce Data Warehouse & ELT Pipeline • ***Kimball Dimensional Modeling & Automated Data Stack***
Technologies: SQL (MSSQL, PostgreSQL), dbt Core, Apache Airflow, Docker, Metabase, Python

- Architected an ELT warehouse implementing **Kimball Star Schema** modeling with surrogate keys, generating transaction facts (**fct_orders**) and SCD Type 2 dimensions.
- Engineered **SCD Type 2 point-in-time joins** to guarantee historical revenue attribution, and resolved sessionization fan-out risks.
- Constructed an automated suite of **213 dbt data tests** (schema integrity, relationship checks, reconciliation), orchestrated via **Apache Airflow** DAGs.
- Developed interactive dashboards in **Metabase**, visualizing conversion funnels, customer lifetime value (LTV), and multi-channel revenue trends.

Proactive 5G Handover Prediction (Diffusion Models) • ***M.Tech Thesis — Predictive Analytics & High-Throughput Telemetry***
Technologies: Python, PyTorch, Diffusion (DDPM/DDIM), GRU, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Developed a machine learning forecasting engine pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** simulating signal degradation paths from multivariate **spatio-temporal** telemetry.
- Addressed extreme **98%/2% class imbalance** using a hybrid **SMOTE + Tomek Links** resampling pipeline, training a Random Forest policy achieving a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • ***Applied NLP & Statistical Text Mining***
Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, Scikit-learn, Pandas, NumPy

- Engineered dual text classification architectures: extracted 768-dim dense **BERT embeddings** paired with **XGBoost**, and constructed a deep hybrid Bi-LSTM / Transformer / 1D-CNN pipeline.
- Applied stratified 80/20 train-test sampling, NFKD text normalization, and early stopping regularization, achieving **99% Precision** and **98% Recall** in synthetic LLM classification.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Microsoft Power BI** (Udemy)
- **SQL & Database Design** (Udemy)
- **Python for Data Science & AI** (Coursera)
- **Celonis Process Mining Foundations** (Credly)
- **IELTS Academic:** 7.5 Band Score